

4.5 Manipulating Power Series

Recall: There are functions (such as $f(x) = e^{-x^2}$) which have no antiderivative of elementary functions.

Can we use the knowledge we have learned so far in Chapter 4 to find power series representations for these kinds of functions?

Motivation: We proved in the previous section that the function $f(x) = \sin(x)$ was equal to its Maclaurin series for all x . That is,

$$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

Use this information to find a power series for $g(x) = \cos(x)$.

The following table is a list of important Maclaurin series that one should know or be able to quickly derive.

Function and Maclaurin Series	Interval of Convergence
$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$	$(-1, 1)$
$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$	$(-\infty, \infty)$
$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$	$(-\infty, \infty)$
$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$	$(-\infty, \infty)$
$\tan^{-1}(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}$	$[-1, 1]$
$\ln(1+x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n+1}$	$(-1, 1]$

There are several things we can do to manipulate given power series representations to obtain new power series representations.

We can Manipulate by:

- 1) Seeing functions as sums of geometric series.
- 2) Taking derivatives and antiderivatives of series & functions.
- 3) Multiplying by polynomials.
- 4) Substituting polynomials (this may change the IOC & ROC).

Example 1: Find a power series representation for $f(x) = \frac{x-2}{(1-x)(1+x)}$

Example 2: Consider the series $\sum_{n=1}^{\infty} nx^n$.

a) Determine a function modeled by the series $\sum_{n=1}^{\infty} nx^n$ and state its radius of convergence.

b) Determine the sum of the series $\sum_{n=1}^{\infty} n\left(\frac{1}{2}\right)^n$.

Example 3: Find the Maclaurin series for $f(x) = x \cos(x)$

Note: Like Example 3, the Taylor series of products of functions with known Taylor series can be found by performing products or quotients of these power series. E.g., $\tan(x) = \sin(x)/\cos(x)$.

Example 4: Find a series that sums to equal $\frac{\pi}{6}$.

- Why would this problem need to be solved differently if we wanted a series that summed to be $\frac{\pi}{3}$?

- Find a series that sums to $\frac{1}{2} \tan^{-1}\left(\frac{1}{2}\right)$.

Recall: The function $f(x) = e^{-x^2}$ has no antiderivative of elementary functions. However, this function is fundamental to understanding statistics and probability.

Example 5: Represent $\int e^{-x^2} dx$ exactly as an infinite series.

Example 6: Use the result of example 5 to approximate the integral $\int_0^1 e^{-x^2} dx$ using a Maclaurin series.

- Determine how close the 2nd partial sum of the series is to the exact value of $\int_0^1 e^{-x^2} dx$.